INCH-POUND

MIL-C-48668E (AR)
AMENDMENT 2
23 August 1994
SUPERSEDING
AMENDMENT 1
25 September 1992

MILITARY SPECIFICATION

CHARGE, PROPELLING, M204
PARTS AND CONTAINER ASSEMBLY

This Amendment forms a part of MIL-C-48868E (AR), dated 30 March 1990, and is approved for use by the U.S. Army Armaments, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 4

3.9 Delete in its entirety and substitute:

"3.9 Homogeneity of material. The material used to manufacture The parts and assemblies shall be uniform in structure or composition throughout. The presence of spots 1/4 inch or larger shall indicate non-homogeneity. The presence of thin membrane-like areas in the nose sections of the part or assembly shall indicate non-homogeneity. The top and bottom halves shall not exhibit non-homogeneities including cracks, splits, cuts, tears, rips or holes when tested as specified in 4.5.7. The Government shall establish visual standards prior to production with the assistance of the contractor."

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* Table I, Closure: Delete: "Composition ...1/ ... 100 2/ ... 3.2 ... 4.5.1", and substitute: "Composition ...1/ 4/ ... 100 2/ ... 3.2 ... 4.5.1."

* Table I, Container Assembly: Delete: "Composition ... 5 2/ ... 3.2 ... 4.5.1", and substitute: "Composition ...1/ ... 5 2/ ... 3.2 ... 4.5.1."

* Table I, Notes: Add Note 4 as follows: "4/ Microphotograph shall serve as evidence of acrylic fibrillation."

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4.4.2.2 Delete Minor 201.

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* 4.4.2.4, Major 101: Delete: "101 Composition ...1/ ... 100 2/ ... 3.2 ... 4.4.3.1/4.5 1", and substitute: "101 Composition ...1/ 4/ ... 100 2/ ... 3.2 ... 4.4.3.1/4.5.1."

* 4.4.2.4, Notes: Add Note 4 as follows: "4/ Microphotograph shall serve as evidence of acrylic fibrillation."

4.4.2.4, Minor 201: Delete "Length 3/", and substitute: "Length (if applicable) 3/."

Add the following:

"205 Diameter (if applicable) 3/ 0.65 3.2 Gage."

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* 4.4.2.5, Major 106: Delete: "106 Composition ... 5 2/ ... 3.2 ... 4.4.3.1/4.5 1", and substitute: "106 Composition ... 4/ ... 5 2/ ... 3.2 ... 4.4.3.1/4.5.1."

* 4.4.2.5, Notes: Add Note 4 as follows: "4/ Microphotograph shall serve as evidence of acrylic fibrillation."

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* 4.4.3.1, Delete in its entirety and substitute:

"4.4.3.1 Composition. The container moldings and closurematerial shall be selected from each lot and tested in accordance with 4.5.1. The container moldings shall be free of silicone. The lot shall be rejected if the sample falls to comply with the requirements."

* 4.4.3.2, Delete in its entirety and substitute:

"4.4.3.2 Stability heat test (134.5 C). The container moldings and closure material shall be selected from each lot and tested In accordance with 4.5.2. The container moldings shall be free of silicone. The lot shall be rejected if the sample fails to comply with the requirements."

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* 4.5.1.3.2.1, Delete in its entirety and substitute:

"4.5.1.3.2.1 Diphenylamine (DPA) - UV Method. Cut sample into 1/4 inch pieces or smaller. Dry to constant weight using a microwave moisture analyzer. Cool the sample to room temperature in a desiccator and weigh 1 gram specimen to the nearest 0.2 mg. Transfer the specimen into a beaker, add 50 ml of 90 percent aqueous methanol, and set the beaker in a warm ultrasonic bath for 10 minutes. Disperse the specimen in the liquid by means of a homogenizer for about 2 minutes or until the specimen is completely disintegrated. Decant the liquid portion into a filtering apparatus fitted with a suitable filter paper, and retain as much of the solid in the beaker. Add a similar amount of 90 percent aqueous methanol into the beaker and redisperse the specimen in the liquid by means of the homogenizer for about 1 minute. Let most of the solid settle, then decant the liquid into the filter. Repeat the operation two more times. Transfer the filtrate into a 250 ml volumetric flask and dilute to the mark with 90 percent aqueous methanol. Shake the flask well and pipet 5 ml of this solution into a 50 ml volumetric flask. Dilute the solution to the mark with 90 percent aqueous methanol and shake well. Measure the absorbance of this solution in the spectrophotometer at 285 nm. Calculate the DPA content of the sample according to the following equation.

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$$\text{Percent DPA} = \frac{(y - a)\ 250}{bw}$$

Where, y = absorbance
a = intercept of calibration curve
b = slope of calibration curve
w = weight of specimen, in grams

* 4.5.1.3.2.2, Delete in its entirety and substitute:

"4.5.1.3.2.2 Calibration procedure. Dry approximately 1 gram of diphenylamine in the microwave moisture analyzer. Weigh approximately 200 mg of dry DPA to the nearest 0.2 mg. Transfer this into a 250 ml volumetric flask. Add 90 percent aqueous methanol and dissolve completely. Dilute to mark and mix well.

Pipet 5 ml of above solution and transfer into a 100 ml volumetric flask. Dilute to mark with 90% aqueous methanol and mix well. This is the standard stock solution. Pipet volumetrically the following aliquots of standard stock solution into 100 ml volumetric flasks and dilute each with 90% aqueous methanol to the mark, then mix well.

Standard	Standard Stock Solution Aliquot (ml)	Total Volume (ml)	Concentration (mg/ml)
1	5	100	W/100,000
2	10	100	2W/100,000
3	15	100	3W/100,000

Where, W = weight of dry DPA, in mg.

Zero the spectrophotometer with fresh 90% aqueous methanol at 285 nm. Rinse the cell with first standard twice, then measure the absorbance. Repeat the same procedure for standards 2 and 3. Perform a linear regression analysis on absorbance and concentration of standard solutions and obtain the slope, intercept, and correlation coefficient of calibration. If the latter value is less than 0.99, rerun the standards in the pectrophotometer."

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* 4.5.1.3.3.1, Delete in its entirety and substitute:

"4.5.1.3.3.1 Nitrocellulose - Schulze-Tiemann Method for Nitrocellulose (NC) Determinations. Calibrate a eudiometer tube prior to assembly of the Schulze-Tiemann apparatus by suspending the tube in an upside down position and filling this with incremental volumes of aqueous ethanol solution from a buret. Take a reading on the audiometer tube for each volume of aqueous ethanol transferred from the buret. Subtract each volume measured with the buret from the corresponding reading on the audiometer tube. Take the average of the differences between the buret and audiometer readings. This average is the audiometer volume correction X.

The value of X is positive if the buret readings are smaller than audiometer readings and vice versa.

Assemble a 250 ml round bottom flask, gas evolution apparatus, audiometer tube, NaOH bath, lead band, heating mantle and leveling bulb as shown in Figure 1.

Calibrate the apparatus with dry standard A.C.S. grade KNO_3. Follow the procedure of analysis below using 0.5 - 0.7 g of standard KNO_3 weighed to the nearest 0.1 mg. Calculate the factor of the apparatus according to the following equation:

$$F = \frac{616.8W\ (273.15 + T)}{(V-X)\ (B-C-P-3.48)}$$

Where, B = barometric pressure (mm Hg)
C = temperature correction (mm Hg), See paragraph 4.5.1.3.3.2
F = S-T Method correction factor
P = vapor pressure of water over 33 percent w/w NaOH at T (mm Hg), see paragraph 4.5.1.3.3.3
T = temperature of condenser water (degrees C)
V = volume of gas in audiometer tube (ml)
W = mass of KNO_3 (g)
X = audiometer volume correction (ml)

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Prepare samples as follows: Cut sample into 1/4 inch pieces or smaller. Dry to constant weight using the microwave moisture analyzer, then store in a desiccator.

Fill the audiometer tube with the NaOH solution by means of the leveling bulb. Weigh between 0.7 and 1.0 g of specimen to the nearest 0.1 mg and transfer this into the round bottom flask. Add 30 ml of deionized (DI) water, heat the water to boiling and purge the flask/gas evolution apparatus assembly with steam for 5 to 10 minutes. Remove the heat for a minute, then add 25 ml of $FeCl_2$ solution, 25 ml concentrated HCl, and 10 ml of DI water.

Place the audiometer tube over the tip of the gas evolution apparatus and allow the reaction to continue for 20-30 minutes or until the bubbling has stopped. Add 30 ml of DI water through the addition funnel in a similar manner and continue heating for 5 to 10 minutes more. Move the audiometer away from the gas evolution apparatus, turn off the heat, and release the vacuum in the flask by opening the addition funnel stopcock. Allow the audiometer to stand for at least 10 minutes, then read the gas volume on the audiometer to the nearest 0.1 ml at equilibrium pressure with the atmosphere using the leveling bulb.

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Calculate the NC content of sample according to the following equation:

$$\text{Percent NC} = \frac{2.246F\,(V-X)(B-C-P-3.48)}{NW\,(273.15+T)}$$

Where, B = barometric pressure (mm Hg)
C = temperature correction (mm Hg), See paragraph 4.5.1.3.3.2
F = S-T Method correction factor
N = percent nitrogen in NC
P = vapor pressure of water over 33 percent w/w NaOH at T (mm Hg), see paragraph 4.5.1.3.3.3
T = temperature of condenser water (degrees C)
V = volume of gas in audiometer tube (ml)
W = mass of sample (g)
X = audiometer volume correction (ml)

Preparation of $FeCl_2$ Solution. Prepare the solution by adding 1000 g of $FeCl_2$.4720, 1200 ml DI water, and 70 ml of concentrated HCl into a 5 liter round bottom flask. Attach a reflux condenser and reflux for 20 minutes. Cool to room temperature. Slowly add 20 grams of iron filings or powder to the cool solution. When the gas evolution stops, filter quickly. Use A.C.S. grade chemicals.

Preparation of NaOH Solution, 33 percent (w/w). Dissolve 330 g of reagent grade NaOH in 670 g of DI water.

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* Delete Figure 1 and substitute a new Figure 1 (as follows).

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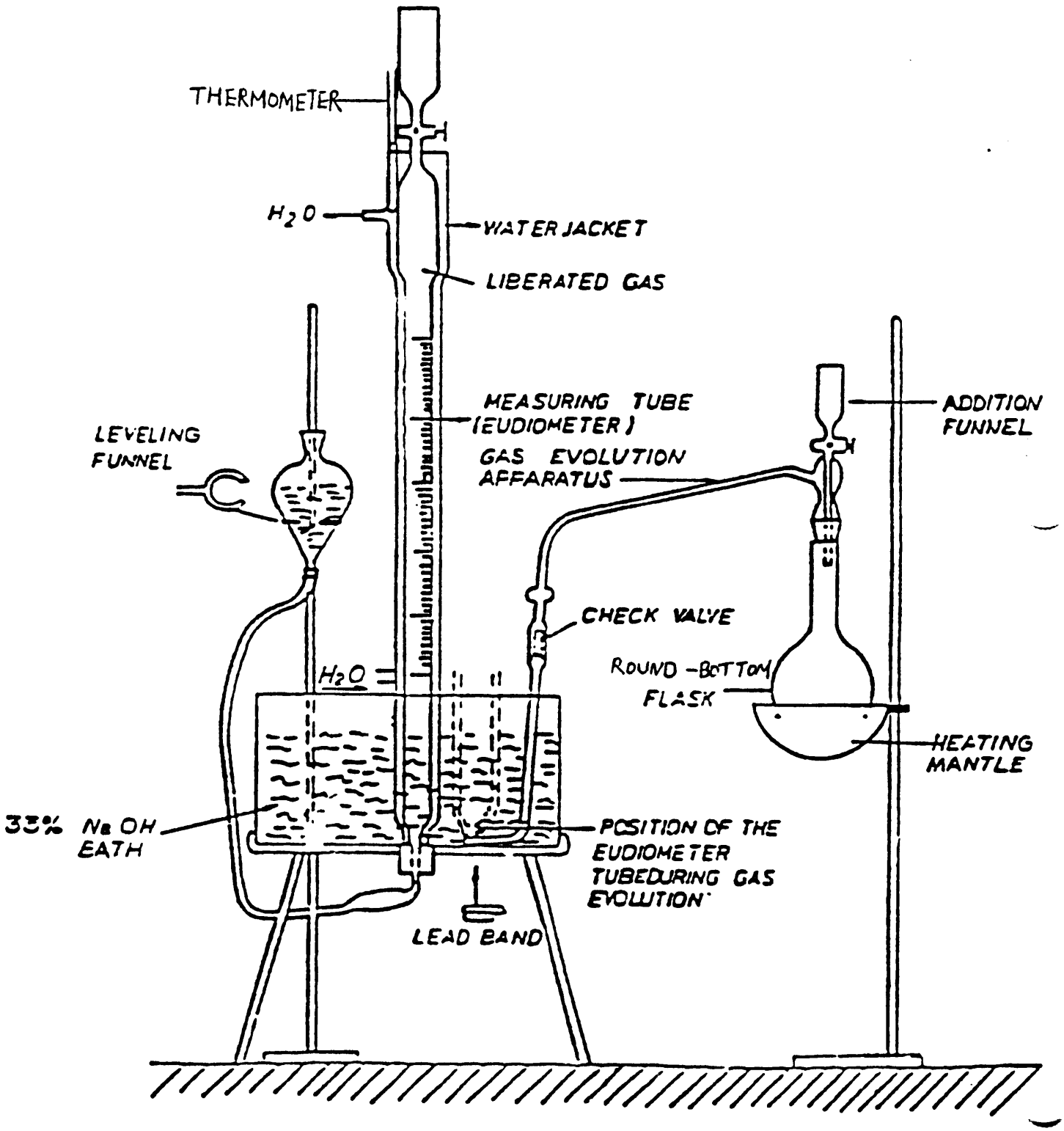


FIGURE 1.
Apparatus for the determination of nitrocellulose by the Schulze-Tiemann method.

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* 4.5.1.3.4.1 Delete in its entirety and substitute:

"4.5.1.3.4.1 Alternate method - preparation of samples. Each sample shall be cut into approximately 1/4 inch squares or circles. Cut and dry, in microwave or oven, approximately 2.5 g from each sample and cool in desiccator. From each sample, approximately 1.0 g test specimen shall be weighed to the nearest 0.2 mg. Test is destructive and sample shall be removed from the lot.

Analytical procedure

Solvent extraction. Place the test specimen in a centrifuge bottle and add about 50 ml of absolute methanol. Completely homogenize the specimen in methanol for approximately 2 minutes with a polytron homogenizer. Turn the homogenizer to low speed, then with a squirt bottle of technical grade acetone, slowly add the acetone down the shaft of the homogenizer until the level reaches about 1/2 inch below the shoulder of the bottle. Stir at slow speed for 2 minutes. Centrifuge the contents in the bottle for 5 minutes and decant the liquid. Repeat the dispersion of the specimen in acetone without methanol, centrifuge the mixture, and decant the clear liquid. Rinse the polytron shaft with acetone and save the rinsing in the centrifuge bottle containing the specimen. Set up a vacuum filter equipped with an appropriate filter paper. Swirl the contents in the bottle, then pour the entire mixture into the filter funnel. Turn on the vacuum. Rinse the shaft with more acetone, catching the rinsing in the bottle, and pouring this into the filter funnel. until all the fibers have been transferred into the filter. When all the liquid has drained into the flask, rinse the sides of the funnel with acetone to collect all the fibers on the filter paper. Leave the vacuum on for about one minute to dry the specimen. Remove the vacuum funnel, then carefully remove the filter paper and fold this over the specimen to insure no loss of material.

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Reagents.	
Sodium hydroxide	400 g solid NaOH dissolved in H_2O and diluted to one liter
Thiosulfate solution	102 g $Na_2S_2O_3.5\,O$ dissolved in $H.O$ and diluted to one liter
Sulfuric acid	93-98 percent H_2SO_4, N_2 free
Hydrochloric acid - standard solution	0.1 N
Sodium hydroxide - standard solution	0.1 N
Boric acid solution	40 g H_3BO_4 dissolved in one liter aqueous solution
Potassium hydrogen phthalate	Reagent grade
Phenolphthalein indicator	1 g phenolphthalein dissolved in 200 ml ethyl alcohol
Kjel-Fast catalyst Kjeldahl digestion catalyst	Mercury–free, such as Kelmate NT

Apparatus. Use a Buchi or CEM Kjel-Fast digester, or equivalent, a pyrex reaction tube or flask, a distillation adapter and a Buchi distillation unit.

Digestion - Using the Buchi Digester. Place the filter paper containing the specimen in a reaction tube and add one packet of Kjeldahl digestion catalyst and 25 ml of concentrated **H** Attach the reaction tube to a manifold (by means of a clamp) which in turn is connected to a water aspirator. Set the reaction tube in the Buchi digester, turn on the latter and allow the reaction to proceed for one hour. Turn off the digester and allow the tube and content to cool to room temperature.

Digestion - Using the Kjel-Fast Digester. Turn on the CEM Kjel-Fast digestion unit and water supply. (The power switch is on the bottom right corner of the back. The water spigot is behind the unit.)

The Kjel-Fast unit will prompt:

S-1 SET TIME -

Press 45 ENTER. Then the prompt will be:

S-1 SET POWER -

Press 100 ENTER

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There are four stages to the program, the unit will prompt you for the next stage after each is entered. The program is as follows:

```
S-1 SET TIME = press  45 ENTER
S-1 SET POWER = press 100 ENTER
S-2 SET TIME  = press  45 ENTER
S-2 SET POWER = press 100 ENTER
S-3 SET TIME  = press 320 ENTER
S-3 SET POWER = press  80 ENTER
S-4 SET TIME  = press 100 ENTER
S-4 SET POWER = press  65 ENTER
```

If a wrong number is pressed, press 0 four times to clear the display. If a wrong number is discovered after ENTER is pressed, press ENTER until the display shows SYSTEM READY then press PROG and start from the beginning. If the program is entered correctly, the system status display will read SYSTEM READY and the time remaining display will read 0740. If any problems occur, check the CEM operation and Service manual for the Kjel-Fast system for more information. With the peristaltic pump in the forward position, lift the switch in the rear of the unit while holding a beaker under the extension tube. Pump Kjel-Fast Catalyst through the system until about 5 mls is pumped through the extension tube. Place the specimen and one packet of Kjeldahl catalyst in the labeled reaction vessel. Add 15 mls concentrated sulfuric acid to the reaction vessel. Place the reaction vessel with insulator on the vessel pedestal inside the microwave cavity. The vessel will catch the Kjel-Fast Catalyst solution that enters the distribution tube. Press start and run the system to completion for about 8 minutes.

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Distillation. Before turning on the distillation unit, open the tap water line and check the $NaOH-Na_2S$ solution and the H_2O reservoirs to make sure that these containers have sufficient amounts of these liquids. NOTE: The $NaOH-Na_2S_2O_3$ solution is a mixture of 10N NaOH solution and a 102 g/1 Na_2S_2O in a 5:1 ratio. Turn the power switch on. Install an empty distillation and a receiving vessel into the unit. Preheat the unit for about 5 minutes. Install the distillation vessel containing the test specimen digest and a receiving vessel containing 100 ml of 40 g/1 boric acid solution. Start the distillation and allow this to continue for about 5 minutes. Lower the receiving vessel and allow the collection tube to drip into the vessel. Set the latter aside for titration. Remove the distillation vessel from the unit and pour the remaining liquid into a waste storage container. Rinse the vessel with water and combine the rinsing with the rest in the waste container.

Analysis. Drain the 10-ml buret twice with standardized 0.1N NCL solution. Standardize the pH meter and electrode using a pH4 and a pH7 buffer solution. Titrate the specimen distillate collected from distillation with the standard 0.1N HCL solution to the pH of the blank solution. Note: The blank solution is prepared at least twice and measured for pH each week. Calculate the percent of acrylic fibers in the specimen as follows:

$$\% \text{ Acrylic Fibers} = \frac{6.05 \times V \times N}{W}$$

Where, V = Volume of standard 0.1 NHCL solution, in ml.
N = Normality of the HCL solution
W = Weight of test specimens, in grams."

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* 4.5.1.3.4.1, under Analysis:

Delete: $\% \text{ Acrylic Fibers} = \frac{6.05 \times V \times N}{W}$

Where, V = Volume of standard 0.1N HCl solution, in ml.
N = Normality of the HCl solution
w = Weight of test specimens, in grams."

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and substitute: "% Acrylic Fibers = $\frac{140 \times V \times N}{KW}$

Where, V = Volume of standard 0.1N HCl solution, in ml.
N = Actual normality of the HCl solution
W = Weight of test specimens, in grams
K = Nitrogen content of acrylic fibers, in %."

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